1 Introduction

The Lotka-Volterra model and the density-dependent predator death model are the mathematical models for the population dynamics of a predator and prey. In this project, we focus on the interactions between snowshoe hares and Canadian bobcats in these two models. To explore snowshoe hares and Canadian bobcats' population dynamics in the two systems.

2 The Lotka-Volterra model

The Lotka-Volterra equations can be written as a system of first-order non-linear ordinary differential equations (ODEs).

$$\begin{cases} \frac{dH}{dt} = rH - aHP \\ \\ \frac{dP}{dt} = eHP - vP \end{cases}$$

From Wikipedia, we know the variable H is the population of snowshoe hares, and the variable P is the population of the bobcat, dH/dt and dP/dt represent the instantaneous growth rates of the two populations, and t represents time. The prey's parameters, r and a, describe, respectively, the maximum prey per capita growth rate, and the effect of the presence of predators on the prey growth rate. The predator parameters, e and v, respectively describe the predator's per capita death rate and the effect of the presence of prey on the predator's growth rate.

2.1 Euler's Method

The assumed initial condition shows that the population of snowshoe hares is set at 100 individuals, and the population of Canadian bobcats is 8. "The estimates, standard errors, and p-values for the four parameters in the selected ODE Model 1 with the parameter cascading method from the real data", which is mentioned in the article "On the selection of ordinary differential equation models with application to predator-prey dynamical models", from that we can get $r = 95$, $a = 4.135$, $e = 1.331$, $v = 50.43$ for the Lotka-Volterra

model. So the Lotka-Volterra equations can be written as,

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \\ \frac{dP}{dt} = 1.331HP - 50.43P \end{cases}$$

I let $\Delta t = 0.01$, Δt represent the size of the time step used in the numerical approximation of a differential equation. Through using Euler's method, we can see the population changes, when the time changes. After 0.01 years, there are 161.96 hares and 14.6136 bobcats. At 0.02 years, there are 218.0723895 hares and 38.74631783 bobcats. At 0.03 years, the population of hares is 75.8542471 and the population of bobcats is 130.9773718.

2.2 Equilibrium Points

Equilibrium points show when snowshoe hares and Canadian bobcats are at this number, the system remains stable, which means their population will not change. For the given system, there are two equilibrium points: $(0, 0)$ and $(37.88880541, 22.97460701)$. The equilibrium point $(0, 0)$ means that both populations are at zero. There are no snowshoe hares and Canadian bobcats, so the snowshoe hares and Canadian bobcats will not grow up in the future. Both of them keep the population at zero. The second equilibrium point $(37.88880541, 22.97460701)$ represents in this system, the number of snowshoe hares born equal to the number of their death naturally plus the number eaten by Canadian bobcats. And the number of Canadian bobcats' deaths naturally equals their birth. So when there are 37.88880541 snowshoe hares and 22.97460701 Canadian bobcats, their population will not change.

2.3 Direction Field and Phase Portrait

From Figure 1, the x-axis represents the population of snowshoe hares and the y-axis represents the population of bobcats. The direction field of snowshoe hares and bobcats shows interactions of their population changes.

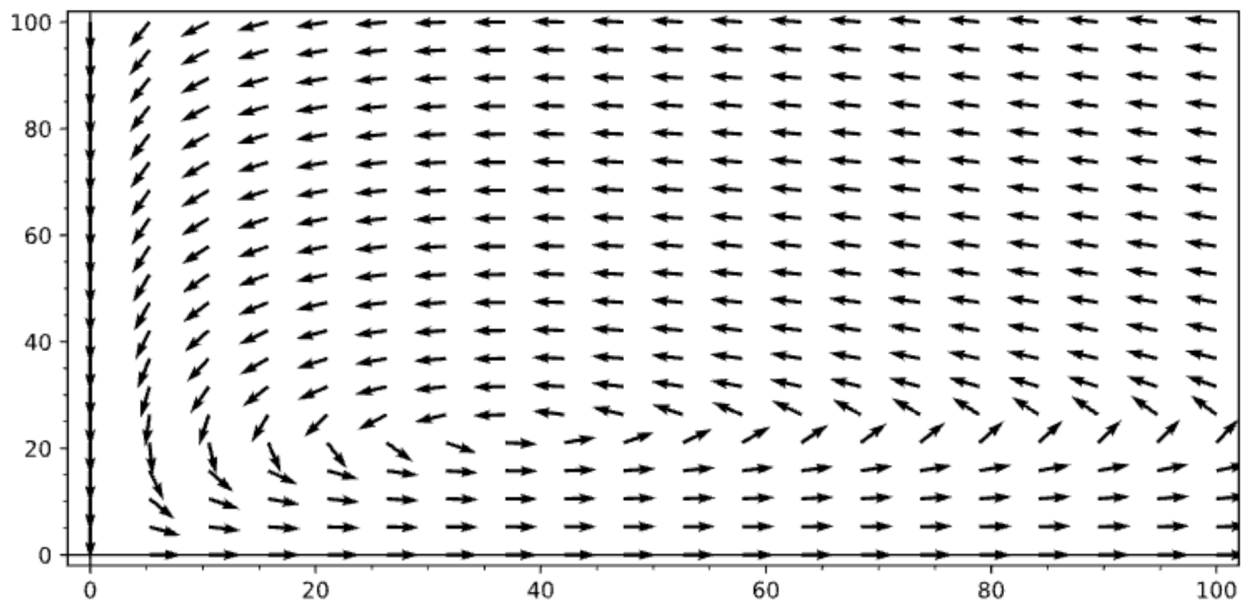

Figure 1: Direction field for snowshoe hares and bobcats

From Figure 2, the phase portrait for snowshoe hares and bobcats helps visualize and direct the behavior of this system. Which can also indicate the stability of the system.

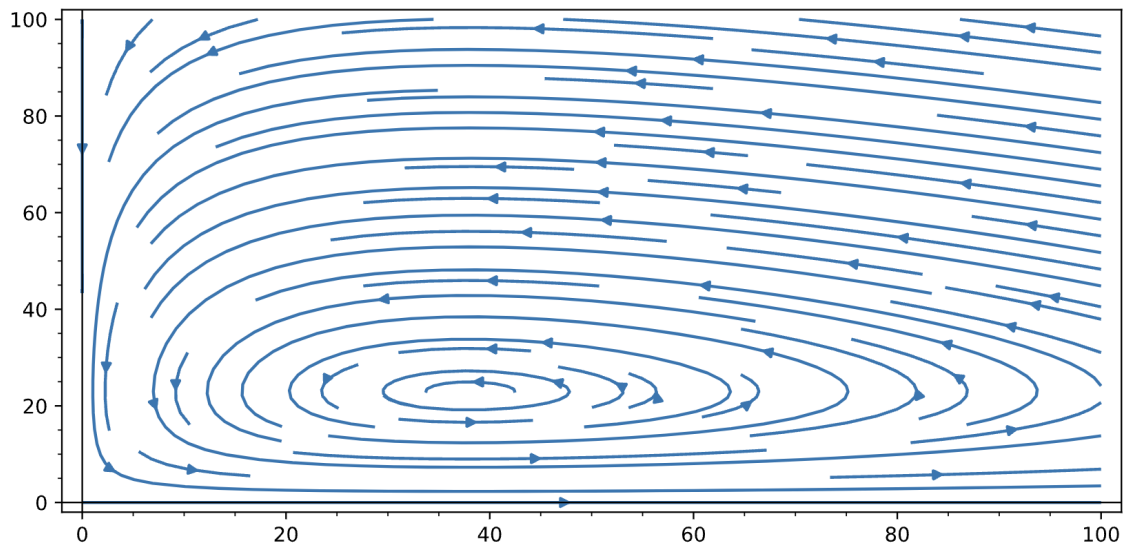

Figure 2: Phase Portrait for snowshoe hares and bobcats

3 The density-dependent predator death model

In the density-dependent predator death model, the death rate of predators depends on the density of both prey and predator populations. It can be written simply as a system of first-order non-linear ordinary differential equations (ODEs).

$$\begin{cases} \frac{dH}{dt} = rH - aHP \\ \frac{dP}{dt} = eHP - vP - bP^2 \end{cases}$$

b represents the rate at which predator death depends on predator density. I assumed the b = 2, so the equations can be written as:

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \frac{dP}{dt} = 1.331HP - 50.43P - 2P^2 \end{cases}$$

The initial condition is: $(H(0), P(0)) = (100, 8)$, $\Delta t = 0.01$

3.1 Euler's Method

The initial condition for hares and bobcats is $(H(0), P(0)) = (100, 8)$, and $\Delta t = 0.01$, Through using Euler's method, we can see the population changes when the time changes. Using Euler's method, after 0.01 years,

there are 161.96 hares and 13.3336 bobcats. At 0.02 years, there are 226.634243 hares and 31.79684392 bobcats.

3.2 Equilibrium Points

Here are three equilibrium points for the density-dependent predator death model, they are $(H, P) = (0, 0)$, $(H, P) = (0, -25.215)$, and $(H, P) = (72.41113, 22.97460701)$. $(H, P) = (0, 0)$means that both populations are at zero. There are no snowshoe hares and Canadian bobcats, so the snowshoe hares and Canadian bobcats will not grow up in the future. Both of them keep the population at zero. $(H, P) = (0, -25.215)$ indicates that there are no snowshoe hares, so bobcats lack prey, and their population decreases. $(H, P) = (72.41113, 22.97460701)$ represents in this system, the number of snowshoe hares born equal to the number of their death naturally plus the number eaten by Canadian bobcats. And the number of Canadian bobcats' deaths naturally equals their birth. So when there are 72.41113 snowshoe hares and 22.97460701 Canadian bobcats, their population will not change.

3.3 Direction Field and Phase Portrait

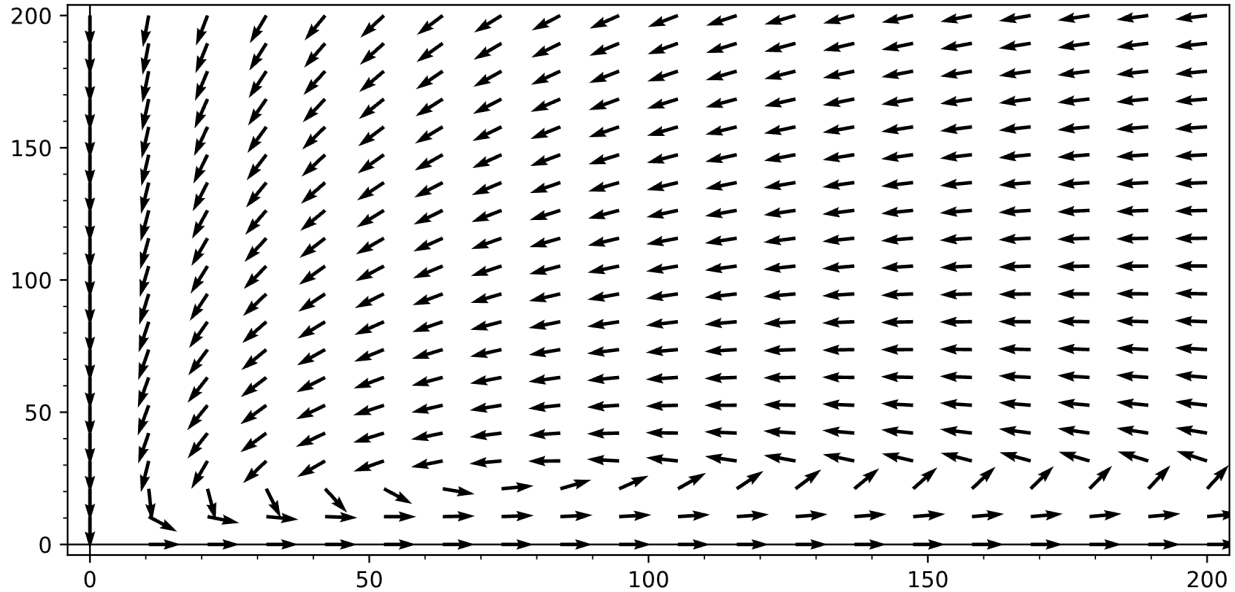

Figure 3: Direction field for snowshoe hares and bobcats

Figure 3 shows how the population of snowshoe hares increases, the bobcats increase because bobcats will have more food. And when hares decrease, the bobcats increase, since hares are eaten by bobcats. The hares decrease, and the bobcats decrease since bobcats don't have more hares be food to live. So there are fewer bobcats live, and the number of hares increases.

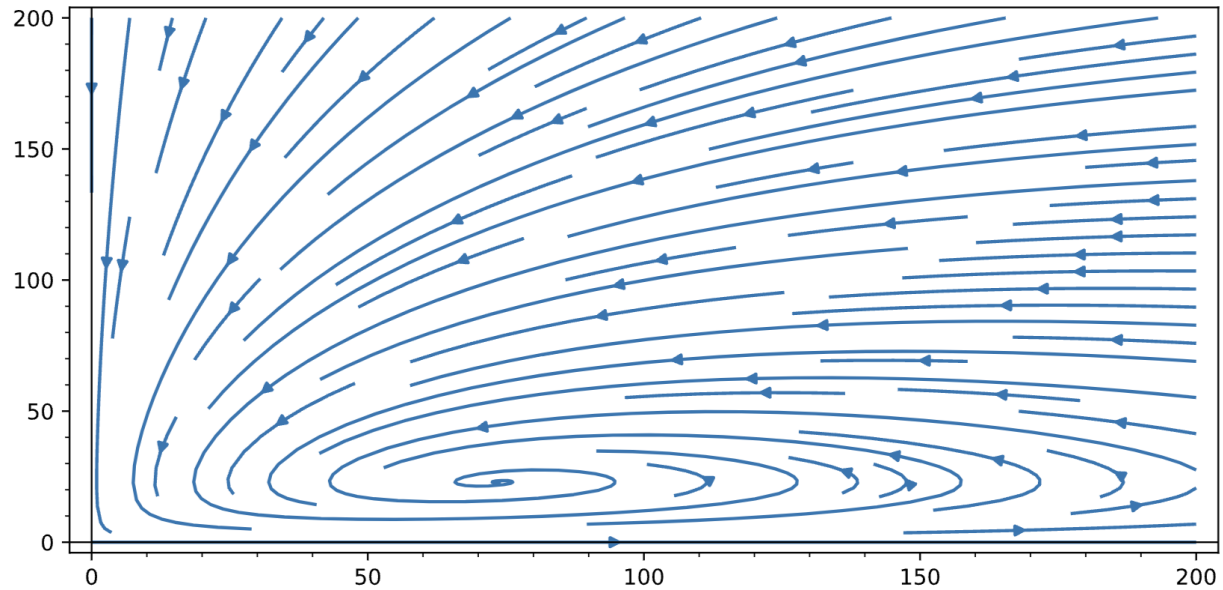

Figure 4: Phase Portrait for snowshoe hares and bobcats

Figure 4 shows the phase portrait for snowshoe hares and bobcats, which helps visualize and direct the behavior of the density-dependent predator death model. The phase portrait can also indicate the stability of the system.

4 Linearize the Lotka-Volterra model

"The estimates, standard errors, and p-values for the four parameters in the selected ODE Model 1 with the parameter cascading method from the real data", which is mentioned in the article "On the selection of ordinary differential equation models with application to predator-prey dynamical models", from that we can get $r = 95$, $a = 4.135$, $e = 1.331$, $v = 50.43$ for the Lotka-Volterra model. The Lotka-Volterra model is written as,

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \frac{dP}{dt} = 1.331HP - 50.43P \end{cases}$$

From the above, I get the equilibrium points for the Lotka-Volterra model,$(H, P) = (0, 0)$, and $(H, P) = (37.88880541, 22.97460701)$. Use $(H, P) = (37.88880541, 22.97460701$ linearize this system to be like,

$$\begin{cases} \frac{dU}{dt} = -156.6702104V \\ \frac{dV}{dt} = 30.57920193U \end{cases}$$

$$\begin{cases} \frac{dU}{dt} \approx -156V \\ \frac{dV}{dt} \approx 31U \end{cases}$$

4.1 Trace-determinant

Given the linear system:

$$\begin{cases} \frac{dU}{dt} = -156V \\ \frac{dV}{dt} = 31U \end{cases}$$

We can represent this system in matrix form as: $A = \begin{bmatrix} 0 & -156 \\ 31 & 0 \end{bmatrix}$

From $A = \begin{bmatrix} 0 & -156 \\ 31 & 0 \end{bmatrix}$, I can get $T = 0, D = 4836$. Therefore, the equilibrium point $(H, P) = (37.88880541, 22.97460701)$ is indeed a center in the HP-plane. The nearby trajectories in the phase plane move in closed orbits or circles around the equilibrium point

4.2 General Solution

By working out the eigenvalues and eigenvectors, using the initial conditions $(H(0), P(0)) = (100, 8)$. I get the general solution for the system. The general solution to this linear system is

$$\mathbf{Y}(t) = \begin{bmatrix} -18e^t \sin\left(\sqrt{4836}t\right) + 100e^t \cos\left(\sqrt{4836}t\right) \\ 18\frac{\sqrt{4836}}{156}e^t \cos\left(\sqrt{4836}t\right) + 100\frac{\sqrt{4836}}{156}e^t \sin\left(\sqrt{4836}t\right) \end{bmatrix}$$

5 Linearize the density-dependent predator death model

In the density-dependent predator death model, the death rate of predators depends on the density of both prey and predator populations. It can be written simply as a system of first-order non-linear ordinary differential equations (ODEs).

$$\begin{cases} \frac{dH}{dt} = rH - aHP \\ \frac{dP}{dt} = eHP - vP - bP^2 \end{cases}$$

b represents the rate at which predator death depends on predator density. I assumed the b = 2, so the equations can be written as:

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \frac{dP}{dt} = 1.331HP - 50.43P - 2P^2 \end{cases}$$

From the above, I get the equilibrium points for the density-dependent predator death model, $(H, P) = (0, 0)$, $(H, P) = (0, -25.215)$, $(H, P) = (72.41113, 22.97460701)$. Use $(H, P) = (72.41113, 22.97460701)$ linearize this system to be like,

$$\begin{cases} \frac{dU}{dt} = -4.135 \times 72.41113V \\ \frac{dV}{dt} = \frac{1.331\times 95}{4.135}U + \frac{4\times 95}{4.135}V \end{cases}$$

which is the linearized system at $(72.41113, 22.97460701)$. To analyze the system better I write this linear system to be like,

$$\begin{cases} \frac{dU}{dt} \approx -299V \\ \frac{dV}{dt} \approx 31U + 92V \end{cases}$$

5.1 Trace-determinant Analysis

The linear system is:

$$\begin{cases} \frac{dU}{dt} \approx -299V \\ \\ \frac{dV}{dt} \approx 31U + 92V \end{cases}$$

We represent this system in matrix form as: $A = \begin{bmatrix} 0 & -299 \\ 31 & 92 \end{bmatrix}$

From $A = \begin{bmatrix} 0 & -299 \\ 31 & 92 \end{bmatrix}$, I can get $T = 92, D = 9269$, $T^2 - 4D = -28612$, $-28612 < 0$,and $T > 0$, $D > 0$, that the equilibrium point $(H, P) = (72.41113, 22.97460701)$ behaves as a spiral source in the HP-plane.

6 Conclusion

Comparing the Lotka-Volterra model with the density-dependent predator death model, I find that in the density-dependent predator death model, there are more snowshoe hares during stable situations. This situation shows up because as the density of bobcats increases, the rate of bobcat death depending on the density will increase rapidly. Therefore, the bobcats' population decreases, and there are more snowshoe hares alive. In the density-dependent predator model, the rate of predator death depends on the density helps the system become more stable.

7 Appendix

7.1 Euler's Method (The Lotka-Volterra model)

The Lotka-Volterra model is,

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \\ \frac{dP}{dt} = 1.331HP - 50.43P \end{cases}$$

and we know the initial condition of hares and bobcats: $(H(0), P(0)) = (100, 8)$

$$\Delta t = 0.01$$

$$\mathbf{Y}_0 = \begin{bmatrix} 100 \\ 8 \end{bmatrix}, \mathbf{F}_0 = \begin{bmatrix} 95H - 4.135HP \\ 1.331HP - 50.43P \end{bmatrix}$$

$\mathbf{Y}_1 = \begin{bmatrix} 100 \\ 8 \end{bmatrix} + 0.01 \times \begin{bmatrix} 95 \times 100 - 4.13 \times 100 \times 8 \\ 1.331 \times 100 \times 8 - 50.43 \times 8 \end{bmatrix} = \begin{bmatrix} 100 \\ 8 \end{bmatrix} + \begin{bmatrix} 61.96 \\ 6.6136 \end{bmatrix} = \begin{bmatrix} 161.96 \\ 14.6136 \end{bmatrix}$

$\mathbf{Y}_2 = \begin{bmatrix} 161.96 \\ 14.6136 \end{bmatrix} + 0.01 \times \begin{bmatrix} 95 \times 161.96 - 4.13 \times 161.96 \times 14.6136 \\ 1.331 \times 161.96 \times 14.6136 - 50.43 \times 14.6136 \end{bmatrix} = \begin{bmatrix} 161.96 \\ 14.6136 \end{bmatrix} + \begin{bmatrix} 56.11238951 \\ 24.13271783 \end{bmatrix} = \begin{bmatrix} 218.0723895 \\ 38.74631783 \end{bmatrix}$

$\mathbf{Y}_3 = \begin{bmatrix} 218.0723895 \\ 38.74631783 \end{bmatrix} + 0.01 \times \begin{bmatrix} 95 \times 218.0723895 - 4.13 \times 218.0723895 \times 38.74631783 \\ 1.331 \times 218.0723895 \times 38.74631783 - 50.43 \times 38.74631783 \end{bmatrix} = \begin{bmatrix} 218.0723895 \\ 38.74631783 \end{bmatrix} + \begin{bmatrix} -142.2181424 \\ 92.92310504 \end{bmatrix} = \begin{bmatrix} 75.8542471 \\ 130.9773718 \end{bmatrix}$

7.2 Equilibrium Points

Here are the equilibrium solutions to the equations:

$$\begin{cases} 95H - 4.135HP = 0 \\ \\ 1.331HP - 50.43P = 0 \end{cases}$$

Then,

$$\begin{cases} H(95 - 4.135P) = 0 \\ \\ P(1.331H - 50.43) = 0 \end{cases}$$

resulting in either

$$\begin{cases} H = 0 \\ \\ P = 0 \end{cases}$$

or,

$$\begin{cases} 95 = 4.135P \\ 1.331H = 50.43 \end{cases}$$

$$\begin{cases} P = 22.97460701 \\ H = 37.88880541 \end{cases}$$

Therefore,

$$\begin{cases} (H, P) = (0, 0) \\ (H, P) = (37.88880541, 22.97460701) \end{cases}$$

7.3 Direction Field and Phase Portrait

```
#Author: Esther Chen
# Define variables and equations
h, p = var ('u,v')
dudt = 95*u - 4.135*u*v
dvdt = 1.331*u*v - 50.4*v
# Create the vector field
f = (dudt, dvdt)
s = sqrt(dudt^2 + dvdt^2)
plot1 = plot_vector_field((dudt/s, dvdt/s), (u, 0, 100), (v,0, 100))
plot1
var('u v')
streamline_plot([95*u-4.135*u*v,1.331*u*v-50.43*v], [u,0,100], [v,0,100])
```

7.4 Euler's Method(The density-dependent predator death model)

The equation system for the density-dependent predator death model is,

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \frac{dP}{dt} = 1.331HP - 50.43P - 2P^2 \end{cases}$$

The initial condition for the hares and bobcats is : $(H(0), P(0)) = (100, 8)$,

$$\Delta t = 0.01$$

$$\mathbf{Y}_0 = \begin{bmatrix} 100 \\ 8 \end{bmatrix}, \mathbf{F}_0 = \begin{bmatrix} 95H - 4.135HP \\ 1.331HP - 50.43P - 2P^2 \end{bmatrix}$$

$\mathbf{Y}_1 = \begin{bmatrix} 100 \\ 8 \end{bmatrix} + 0.01 \times \begin{bmatrix} 95\times100 - 4.13\times100\times8 \\ 1.331\times100\times8 - 50.43\times8 - 2\times64 \end{bmatrix} = \begin{bmatrix} 100 \\ 8 \end{bmatrix} + \begin{bmatrix} 61.96 \\ 5.3336 \end{bmatrix} = \begin{bmatrix} 161.96 \\ 13.3336 \end{bmatrix}$

$\mathbf{Y}_2 = \begin{bmatrix} 161.96 \\ 13.3336 \end{bmatrix} + 0.01 \times \begin{bmatrix} 95\times161.96 - 4.13\times161.96\times13.3336 \\ 1.331\times161.96\times13.3336 - 50.43\times13.3336 - 2\times13.3336\times13.3336 \end{bmatrix} = \begin{bmatrix} 161.96 \\ 13.3336 \end{bmatrix} + \begin{bmatrix} 64.67424295 \\ 18.46324392 \end{bmatrix} = \begin{bmatrix} 226.634243 \\ 31.79684392 \end{bmatrix}$

7.5 Equilibrium Points

Here are the equilibrium solutions to the equations:

$$\begin{cases} 95H - 4.135HP = 0 \\ 1.331HP - 50.43P - 2P^2 = 0 \end{cases}$$

Then,

$$\begin{cases} H(95 - 4.135P) = 0 \\ P(1.331H - 50.43 - 2P) = 0 \end{cases}$$

resulting in either

$$\begin{cases} H = 0 \\ P = 0 \quad \text{or} \quad P = -25.215 \end{cases}$$

or,

$$\begin{cases} 95 = 4.135P \\ P(1.331H - 50.43 - 2P) = 0 \end{cases}$$

$$\begin{cases} P = \frac{95}{4.135} = 22.97460701 \\ 1.331H = 50.43 + 2 \times \frac{95}{4.135} \end{cases}$$

$$\begin{cases} P = 22.97460701 \\ H = 72.41113 \end{cases}$$

Therefore,

$$\begin{cases} (H, P) = (0, 0) \\ (H, P) = (0, -25.215) \\ (H, P) = (72.41113, 22.97460701) \end{cases}$$

7.6 Direction Field and Phase Portrait

```
#Author: Esther Chen
# Define variables and equations
h, p = var ('u,v')
dudt = 95*u - 4.135*u*v
dvdt = 1.331*u*v - 50.4*v-2*v^2
# Create the vector field
f = (dudt, dvdt)
s = sqrt(dudt^2 + dvdt^2)
plot1 = plot_vector_field((dudt/s, dvdt/s), (u, 0, 200), (v,0, 200))
plot1
var('u v')
streamline_plot([95*u-4.135*u*v,1.331*u*v-50.43*v-2*v^2], [u,0,200], [v,0,200])
```

7.7 Linearize the Lotka-Volterra model

The Lotka-Volterra model is to be written as,

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \frac{dP}{dt} = 1.331HP - 50.43P \end{cases}$$

From the above, we get the equilibrium points,

$$\begin{cases} (H, P) = (0, 0) \\ (H, P) = (37.88880541, 22.97460701) \end{cases}$$

To linearize the system, using

$$\begin{cases} U = H - \frac{50.43}{1.331} \\ V = P - \frac{95}{4.135} \end{cases}$$

and,

$$\begin{cases} \frac{dU}{dt} = \frac{dH}{dt} \\ \frac{dV}{dt} = \frac{dP}{dt} \end{cases}$$

$$\frac{dU}{dt} = 95(U + \frac{50.43}{1.331}) - 4.135(U + \frac{50.43}{1.331})(V + \frac{95}{4.135})$$
$$\frac{dU}{dt} = 95U + 95 \times \frac{50.43}{1.331} - 4.135 \times (UV + \frac{50.43}{1.331}V + \frac{95}{4.135}U + \frac{50.43}{1.331} \times \frac{95}{4.135})$$
$$\frac{dU}{dt} = -4.135UV - 156.6702104V$$

and

$$\frac{dV}{dt} = 1.331 \times (U + \frac{50.43}{1.331}) \times (V + \frac{95}{4.135}) - 50.43 \times (V + \frac{95}{4.135})$$
$$\frac{dV}{dt} = 1.331 \times (UV + \frac{50.43}{1.331}V + \frac{95}{4.135}U + \frac{50.43}{1.331} \times \frac{95}{4.135}) - 50.43V - 50.43 \times \frac{95}{4.135}$$
$$\frac{dV}{dt} = 1.331UV + 30.57920193U$$

Since it is a linear system, the final system is written as,

$$\begin{cases} \frac{dU}{dt} = -156.6702104V \\ \frac{dV}{dt} = 30.57920193U \end{cases}$$

which is the linearized system at (37.88880541,22.97460701)

To analyze the system better, I write this linear system to be like,

$$\begin{cases} \frac{dU}{dt} \approx -156V \\ \frac{dV}{dt} \approx 31U \end{cases}$$

7.8 Trace-determinant

The linear system is,

$$\begin{cases} \frac{dU}{dt} = -156V \\ \frac{dV}{dt} = 31U \end{cases}$$

So, $A = \begin{bmatrix} 0 & -156 \\ 31 & 0 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} -\lambda & -156 \\ 31 & -\lambda \end{bmatrix}$

$$\det(A - \lambda I) = \lambda^2 + 4836$$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 + 4836 = 0$$

$$\lambda = \pm\sqrt{4836}i$$

The eigenvalues are purely imaginary, the system is undamped. The equilibrium point $(H, P) = (37.88880541, 22.97460701)$ is a center in the HP-plane.

7.9 General Solution

$$\lambda = \sqrt{4836}i$$

$$A - \lambda I = \begin{bmatrix} -\sqrt{4836}i & -156 \\ 31 & -\sqrt{4836}i \end{bmatrix}$$

$$(A - \lambda I)\mathbf{V} = 0$$

$\begin{bmatrix} -\sqrt{4836}i & -156 \\ 31 & -\sqrt{4836}i \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, I get the function $\begin{cases} -\sqrt{4836}i \times a - 156b = 0 \\ 31a - -\sqrt{4836}i \times b = 0 \end{cases}$, so $\begin{cases} a = i \\ b = \frac{\sqrt{4836}}{156} \end{cases}$

Therefore, $\mathbf{V} = \begin{bmatrix} a = i \\ b = \frac{\sqrt{4836}}{156} \end{bmatrix}$, $\mathbf{Z} = e^{\sqrt{4836}it} \begin{bmatrix} i \\ \frac{\sqrt{4836}}{156} \end{bmatrix}$

$$\mathbf{Z} = (\cos\left(\sqrt{4836}t\right) + i\sin\left(\sqrt{4836}t\right)) \begin{bmatrix} i \\ \frac{\sqrt{4836}}{156} \end{bmatrix}$$

$$= \begin{bmatrix} -\sin\left(\sqrt{4836}t\right) \\ \frac{\sqrt{4836}}{156}\cos\left(\sqrt{4836}t\right) \end{bmatrix} + i \begin{bmatrix} \cos\left(\sqrt{4836}t\right) \\ \frac{\sqrt{4836}}{156}\sin\left(\sqrt{4836}t\right) \end{bmatrix}$$

$$\mathbf{Y}(t) = \begin{bmatrix} -k_1\sin\left(\sqrt{4836}t\right) + k_2\cos\left(\sqrt{4836}t\right) \\ k_1\frac{\sqrt{4836}}{156}\cos\left(\sqrt{4836}t\right) + k_2\frac{\sqrt{4836}}{156}\sin\left(\sqrt{4836}t\right) \end{bmatrix}$$

Since the initial condition is $(H(0), P(0)) = (100, 8)$, put $(H(0), P(0)) = (100, 8)$ into the general solution, get $k_2 = 100$, $k_1 \approx 18$. Therefore the general solution is,

$$\mathbf{Y}(t) = \begin{bmatrix} -18\sin\left(\sqrt{4836}t\right) + 100\cos\left(\sqrt{4836}t\right) \\ 18\frac{\sqrt{4836}}{156}\cos\left(\sqrt{4836}t\right) + 100\frac{\sqrt{4836}}{156}\sin\left(\sqrt{4836}t\right) \end{bmatrix}$$

7.10 Linearize the density-dependent predator death model

From the above, the density-dependent predator death model can be written as,

$$\begin{cases} \frac{dH}{dt} = 95H - 4.135HP \\ \frac{dP}{dt} = 1.331HP - 50.43P - 2P^2 \end{cases}$$

I also get the equilibrium points, $(H, P) = (\frac{50.43}{1.331} + \frac{190}{5.503658}, \frac{95}{4.135})$

To linearize the system, using

$$\begin{cases} U = H - (\frac{50.43}{1.331} + \frac{190}{5.503658}) \\ V = P - \frac{95}{4.135} \end{cases}$$

and,

$$\begin{cases} \frac{dU}{dt} = \frac{dH}{dt} \\ \frac{dV}{dt} = \frac{dP}{dt} \end{cases}$$

Therefore, the function can be written like this,

$$\frac{dU}{dt} = 95(U + \frac{50.43}{1.331} + \frac{190}{5.503658}) - 4.135(U + \frac{50.43}{1.331} + \frac{190}{5.503658})(V + \frac{95}{4.135})$$

$$\frac{dU}{dt} = 95U + 95\times(\frac{50.43}{1.331} + \frac{190}{5.503658}) - 4.135\times(UV + (\frac{50.43}{1.331} + \frac{190}{5.503658})V + \frac{95}{4.135}U + (\frac{50.43}{1.331} + \frac{190}{5.503658})\times\frac{95}{4.135})$$

$$\frac{dU}{dt} = -4.135UV - 4.135 \times 72.41113V$$

and

$$\frac{dV}{dt} = 1.331 \times (U + (\frac{50.43}{1.331} + \frac{190}{5.503658})) \times (V + \frac{95}{4.135}) - 50.43 \times (V + \frac{95}{4.135}) - 2(V + \frac{95}{4.135})^2$$

$$\frac{dV}{dt} = 1.331UV + \frac{1.331 \times 95}{4.135}U - 2V^2 + \frac{4 \times 95}{4.135}V$$

Since there is a linear system, the system should be written as,

$$\begin{cases} \frac{dU}{dt} = -4.135 \times 72.41113V \\ \frac{dV}{dt} = \frac{1.331\times 95}{4.135}U + \frac{4\times 95}{4.135}V \end{cases}$$

which is the linearized system at (72.41113,22.97460701)

To analyze the system better I write this linear system to be like,

$$\begin{cases} \frac{dU}{dt} \approx -299V \\ \frac{dV}{dt} \approx 31U + 92V \end{cases}$$

7.11 Trace-determinant

The linear system is,

$$\begin{cases} \frac{dU}{dt} \approx -299V \\ \frac{dV}{dt} \approx 31U + 92V \end{cases}$$

So, $A = \begin{bmatrix} 0 & -299 \\ 31 & 92 \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $A - \lambda I = \begin{bmatrix} -\lambda & -299 \\ 31 & 92-\lambda \end{bmatrix}$

$$\det(A - \lambda I) = (92 - \lambda) \times (-\lambda) + 299 \times 31$$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 - 92\lambda + 299 \times 31 = 0$$

$$\lambda = \frac{92 \pm \sqrt{92^2 - 4 \times 299 \times 31}}{2}$$

$$\lambda = \frac{92 \pm 169i}{2}$$

$$\lambda = 46 \pm \frac{169}{2}i$$

References

[1] Xinyu Zhang, Jiguo Cao, and Raymond J. Carroll. "On the selection of ordinary differential equation models with application to predator-prey dynamical models" *Biometrics*, vol. 71, 2014, pp. 131-138, https://doi.org/10.1111/biom.12243. Accessed 2, Mar. 2022.

I got the Lotka–Volterra model and the density-dependent functional response model from this article. Understood the implications of each model. From "The estimates, standard errors, and p-values for the four parameters in the selected ODE Model 1 with the parameter cascading method from the real data." in the article, I used the estimates of the maximum prey per capita growth rate, the effect of the presence of predators on the prey growth rate, the predator's per capita death rate, and the effect of the presence of prey on the predator's growth rate for two models obtained in this article. They are $r = 95$, $a = 4.135$, $e = 1.331$, $v = 50.43$.

[2] Wikimedia Foundation. (2024, April 22). Lotka–Volterra equations. Wikipedia. https://en.wikipedia.org/wiki/Lotka

I used Wikipedia to explain every parameter's meaning in the Lotka-Volterra model.

Peer Review/Writing Center Reflection

1. With whom did you meet for peer review or at the Writing Center? When (date and time) and for how long did you visit with your peer reviewer/Writing Center consultant?

   I met Avalynn Morse at the Writing Center. I visit with the Writing Center consultant on Sunday, April 14, 2024, between 9:30 pm and 10:00 pm.

2. What did you discuss with your peer reviewer/Writing Center consultant?

   We discussed my paper. Does it flow in sentences and can someone who is not a math major understand what I am saying?

3. How did you use the suggestions from your peer reviewer/Writing Center consultant in the writing process?

   She offered suggestions for some words spelling wrong and changed the order for my appendix and conclusion.

4. What did you find helpful by having a peer reviewer/Writing Center consultant read your project?

   Having a Writing Center consultant review my project helps me ensure that it is understandable to others.

5. Why do you think this project is important in a liberal arts education?

   It helps enhance my writing skills and improve my ability to utilize scholarly resources.